

Circuit Theory

Chapter 5

Operational Amplifier

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Operational Amplifier - Chapter 5

5.1 What is an Op Amp?

5.2 Ideal Op Amp

5.3 Configuration of Op Amp

5.4 Cascaded Op Amp

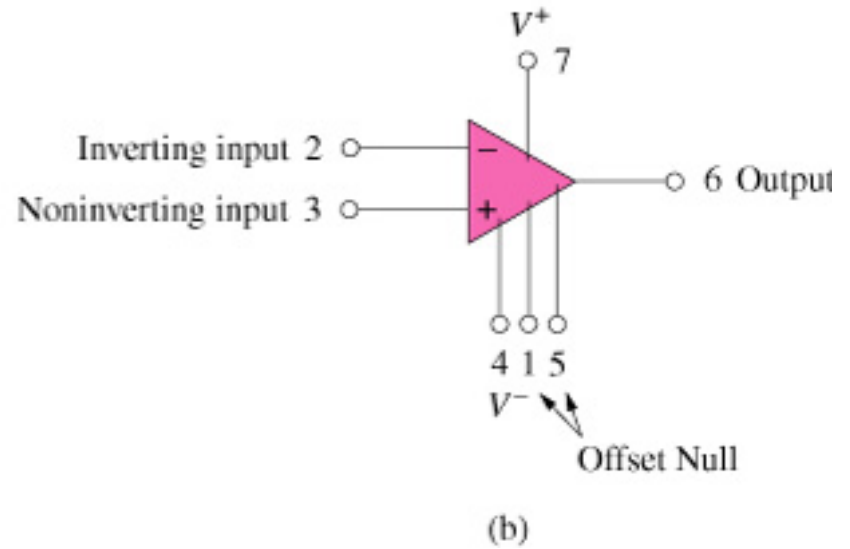
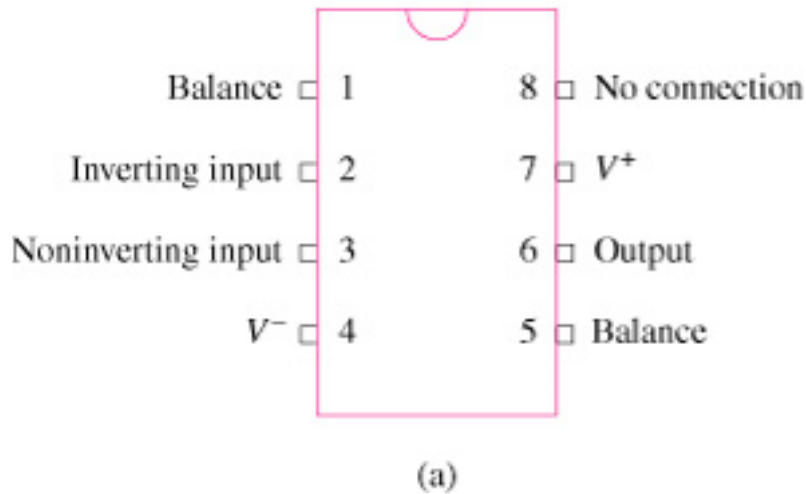
5.5 Application

- Digital-to Analog Converter

5.1 What is an Op Amp (1)

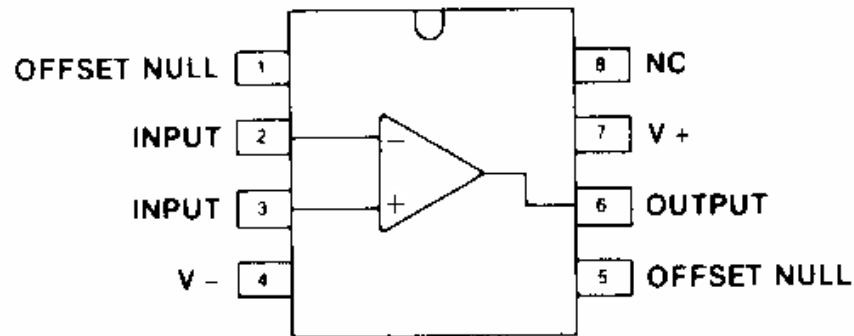
- It is an electronic unit that behaves like a voltage-controlled voltage source.
- It is an active circuit element designed to perform mathematical operations of ***addition, subtraction, multiplication, division, differentiation*** and ***integration***.

5.1 What is an Op Amp (2)

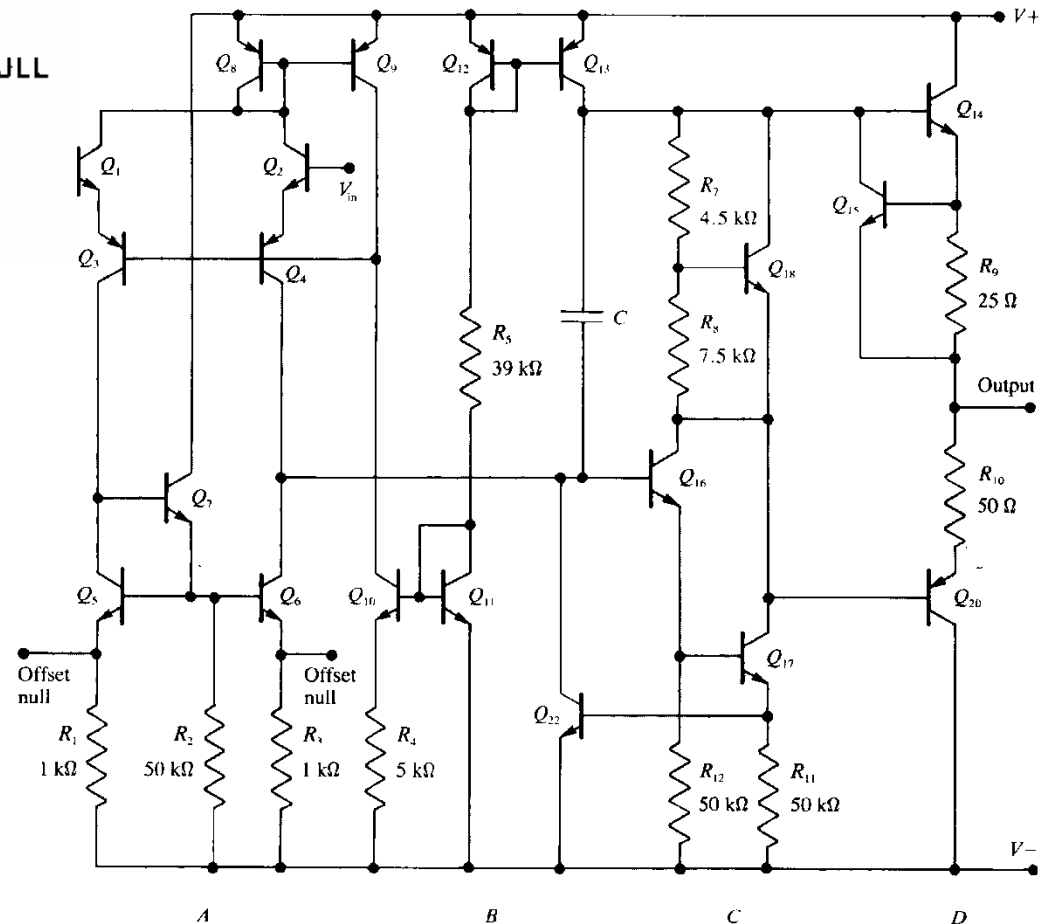


A typical op amp: (a) pin configuration, (b) circuit symbol

5.1 What is an Op Amp (3)

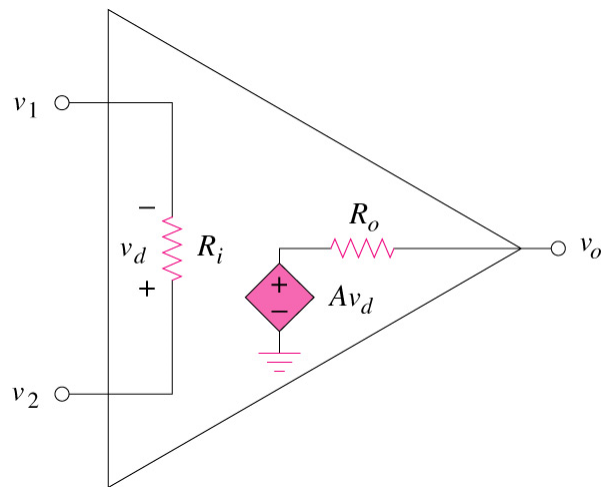


741 op amp

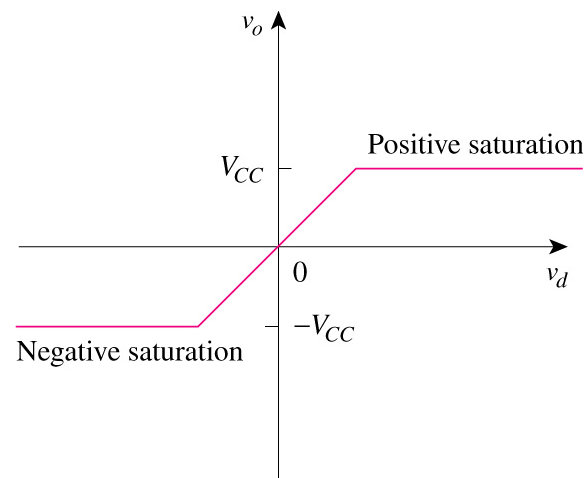


5.1 What is an Op Amp (4)

**The equivalent circuit
Of the non-ideal op amp**



**Op Amp output:
 v_o as a function of V_d**



$$v_d = v_2 - v_1; \quad v_o = Av_d = A(v_2 - v_1)$$

5.1 What is an Op Amp (5)

Typical ranges for op amp parameters

Parameter	Typical range	Ideal values
Open-loop gain, A	10^5 to $10^8 \Omega$	∞
Input resistance, R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance, R_o	10 to 100 Ω	0 Ω
Supply voltage, V_{CC}	5 to 24 V	

5.1 What is an Op Amp (6)

For the op amp circuit of Fig. 5.44, the op amp has an open-loop gain of 100,000, an input resistance of 10 k Ω , and an output resistance of 100 Ω . Find the voltage gain v_o/v_i using the nonideal model of the op amp.

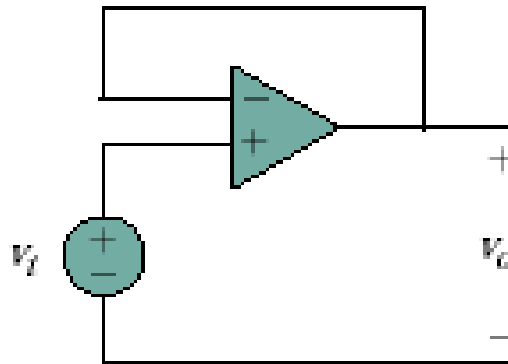
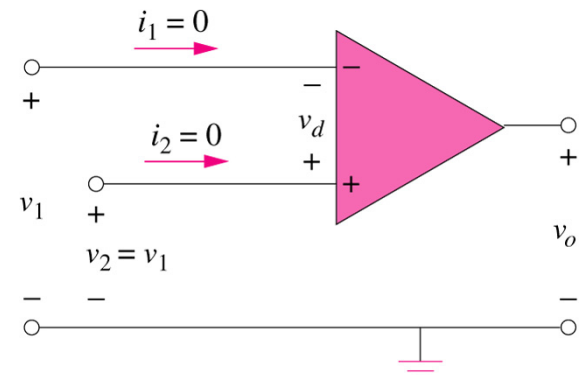


Figure 5.43 For Prob. 5.5.

5.2 Ideal Op Amp (1)

An ideal op amp has the following characteristics:

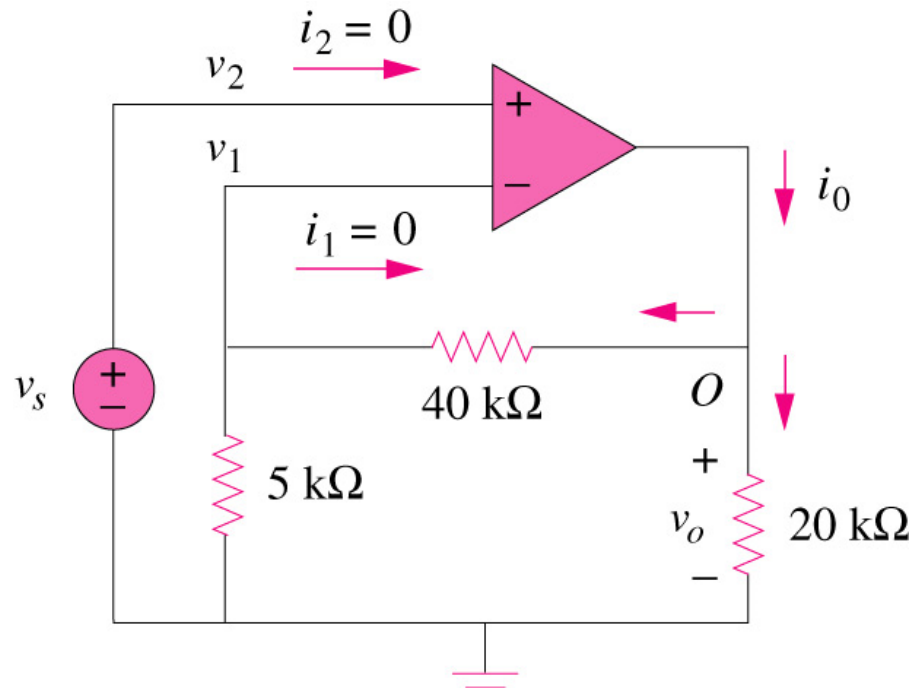
1. Infinite open-loop gain, $A \approx \infty$
2. Infinite input resistance, $R_i \approx \infty$
3. Zero output resistance, $R_o \approx 0$



5.2 Ideal Op Amp (2)

Example 1:

$v_s = 1$ v. Determine the value of i_o .

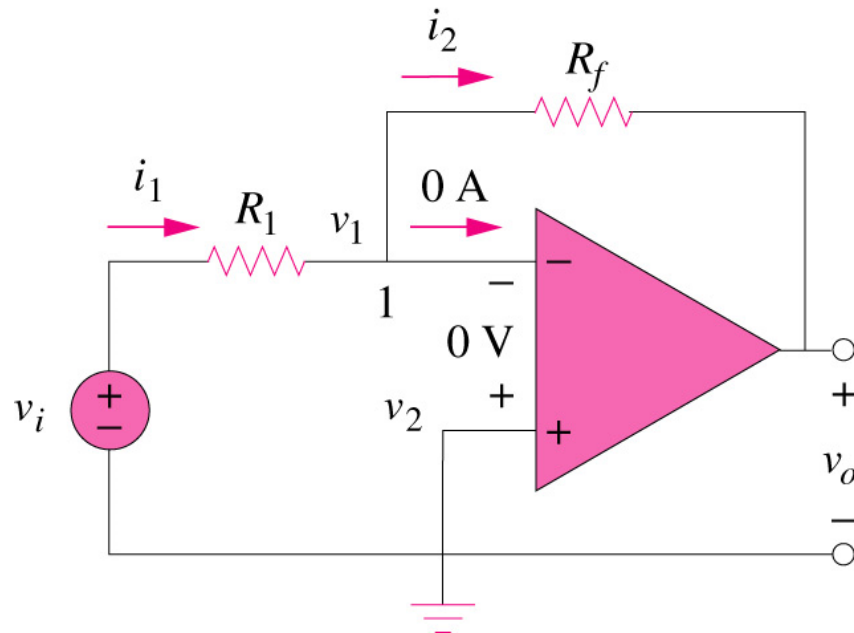


*Refer to in-class illustration, textbook

Ans: 0.65mA

5.3 Configuration of Op amp (1)

- Inverting amplifier reverses the polarity of the input signal while amplifying it

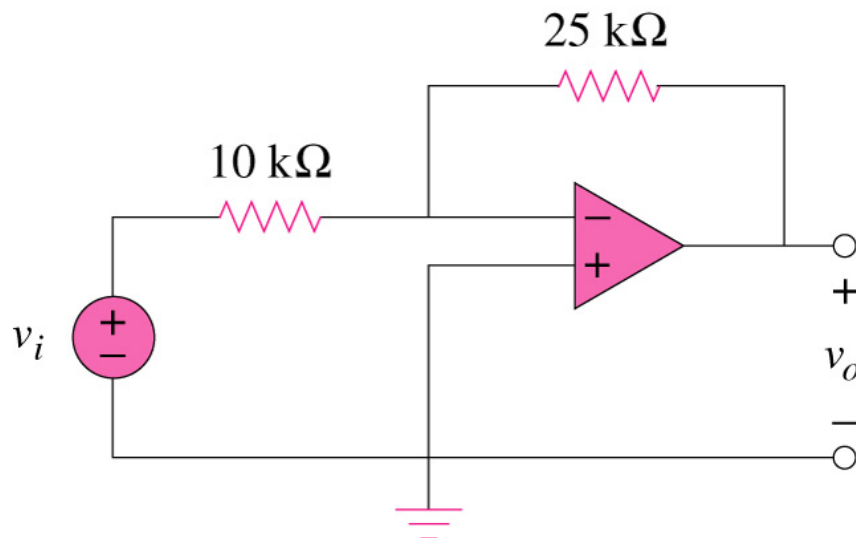


$$v_o = -\frac{R_f}{R_1} v_i$$

5.3 Configuration of Op amp (2)

Example 2

Refer to the op amp below. If $v_i = 0.5\text{V}$, calculate:
(a) the output voltage, v_o and (b) the current in the $10\text{k}\Omega$ resistor.



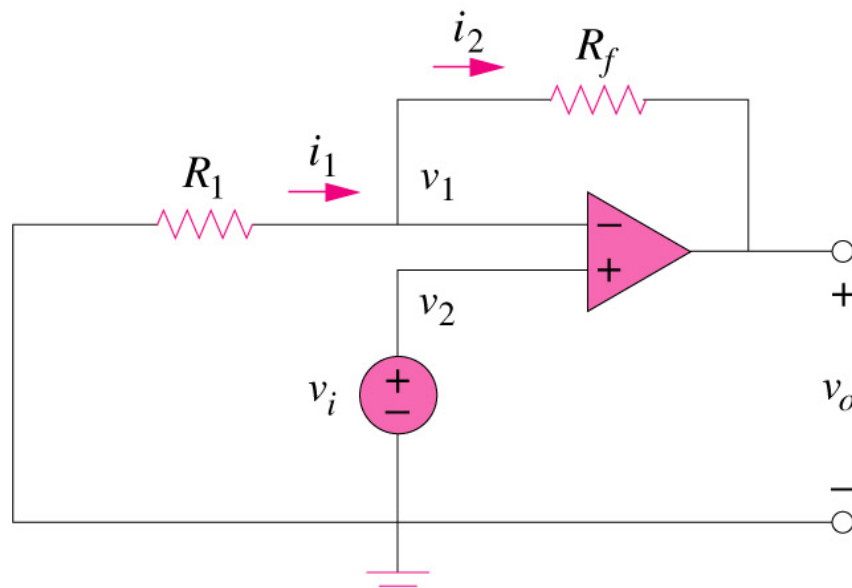
Ans:

(a) -1.25V ; (b) $50\mu\text{A}$

*Refer to in-class illustration, textbook

5.3 Configuration of Op amp (3)

- Non-inverting amplifier is designed to produce positive voltage gain

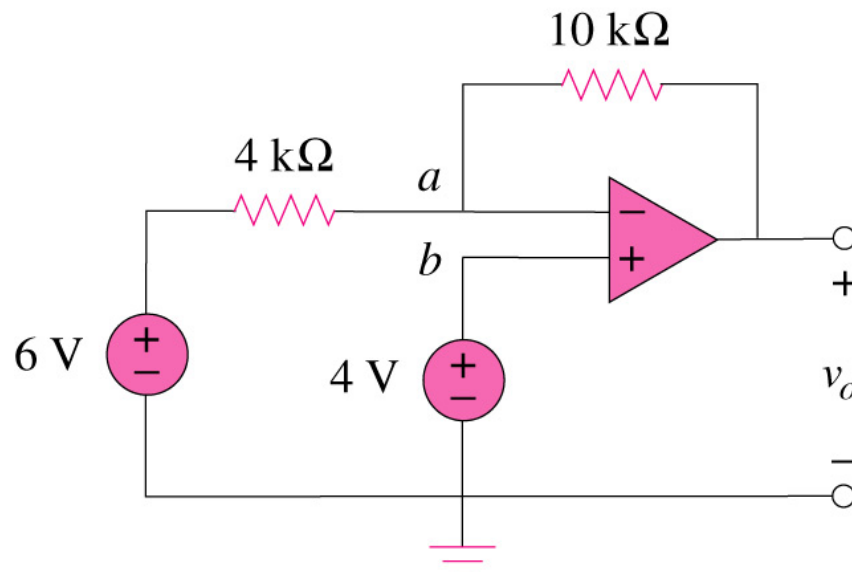


$$v_o = \left(1 + \frac{R_f}{R_1} \right) v_i$$

5.3 Configuration of Op amp (4)

Example 3

For the op amp shown below, calculate the output voltage v_o .

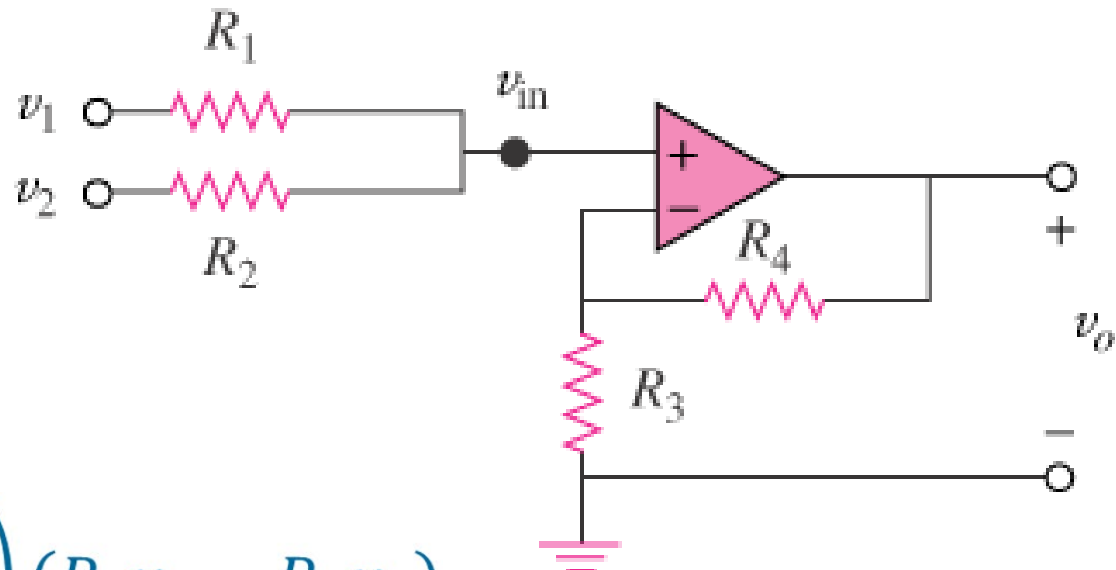


*Refer to in-class illustration, textbook

Ans: -1V

5.3 Configuration of Op amp (4)

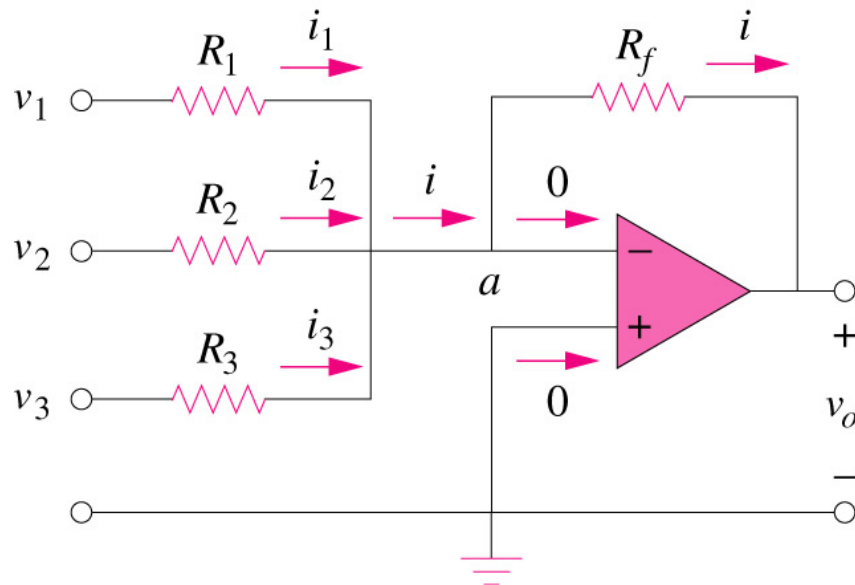
Given the op amp circuit shown below, express v_o in terms of v_1 and v_2 .



$$v_o = \left(1 + \frac{R_4}{R_3}\right) \left(\frac{1}{R_1 + R_2}\right) (R_1 v_2 - R_2 v_1)$$

5.3 Configuration of Op amp (5)

- Summing Amplifier is an op amp circuit that combines several inputs and produces an output that is the weighted sum of the inputs.

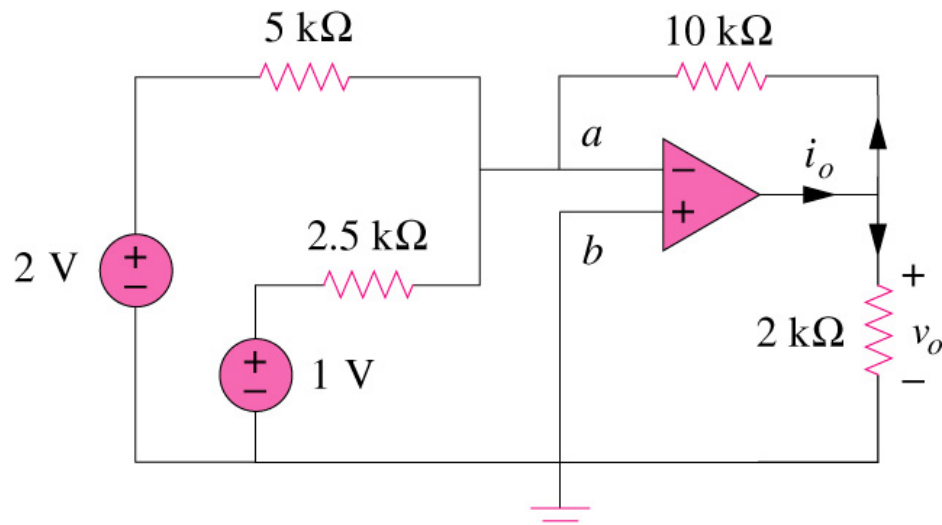


$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$

5.3 Configuration of Op amp (6)

Example 4

Calculate v_o and i_o in the op amp circuit shown below.

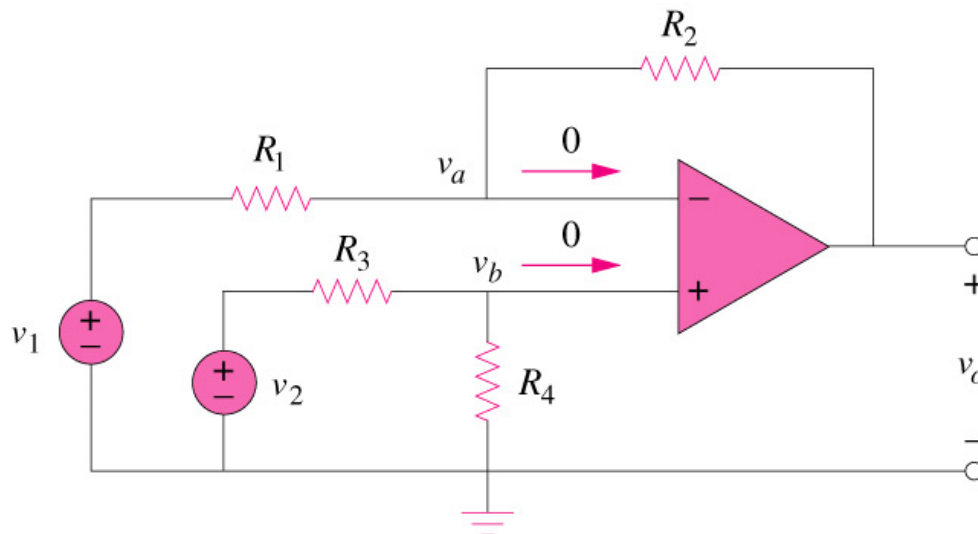


*Refer to in-class illustration, textbook

Ans: -8 V , -4.8 mA

5.3 Configuration of Op amp (7)

- Difference amplifier is a device that amplifies the difference between two inputs but rejects any signals common to the two inputs.

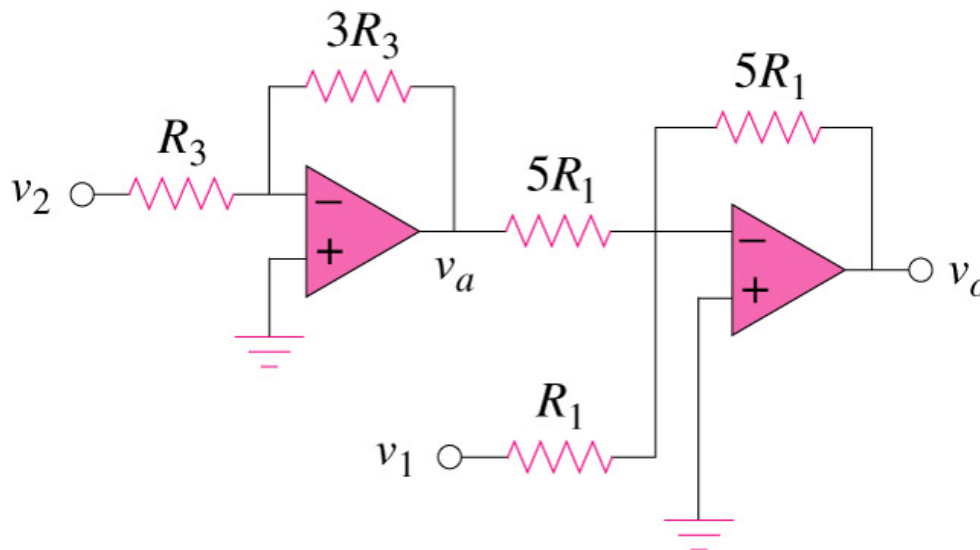


$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1 \Rightarrow v_o = v_2 - v_1, \text{ if } \frac{R_1}{R_2} = \frac{R_3}{R_4} = 1$$

5.3 Configuration of Op amp (6)

Example 5

Determine R_1 , R_2 , R_3 and R_4 so that $v_o = -5v_1 + 3v_2$ for the circuit shown below.



Ans:

$$R_1 = 10\text{k}\Omega$$

$$R_2 = 50\text{k}\Omega$$

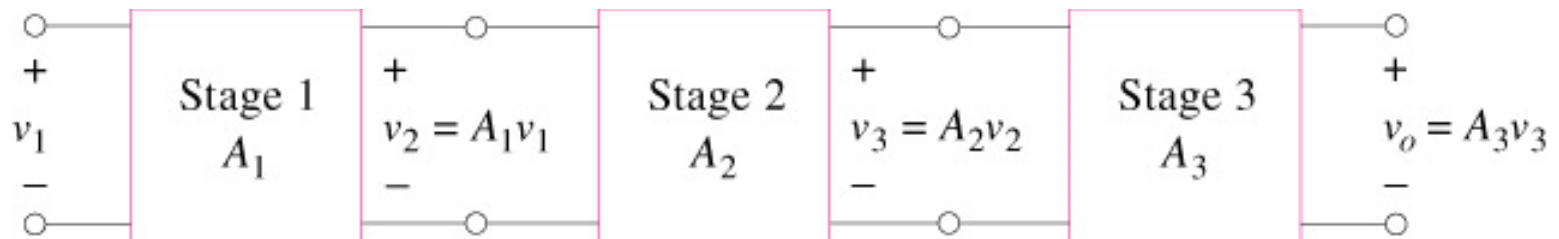
$$R_3 = 20\text{k}\Omega$$

$$R_4 = 20\text{k}\Omega$$

*Refer to in-class illustration, textbook

5.4 Cascaded Op Amp (1)

- It is a head-to-tail arrangement of two or more op amp circuits such that the output to one is the input of the next.

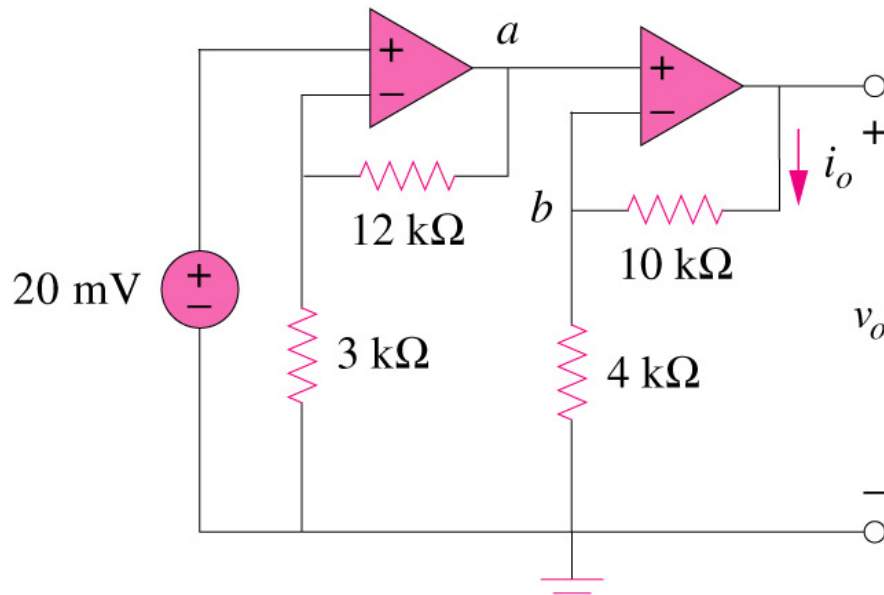


$$v_o = A_1 A_2 A_3 v_1$$

5.4 Cascaded Op Amp (2)

Example 6

Find v_o and i_o in the circuit shown below.



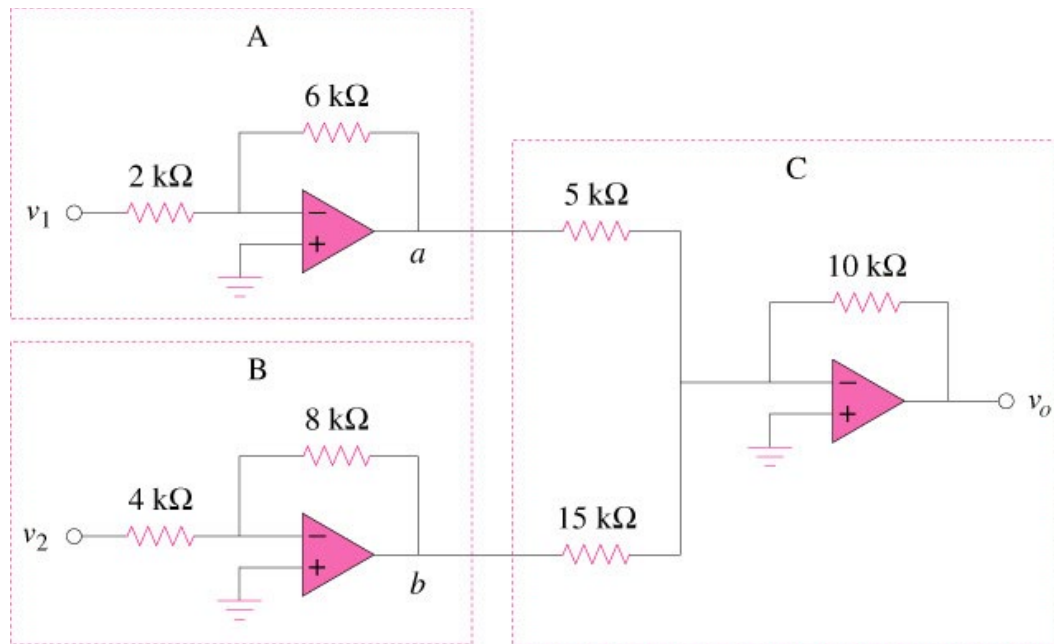
*Refer to in-class illustration, textbook

Ans: 350mV, 25μA

5.4 Cascaded Op Amp (3)

Example 7

If $v_1 = 1V$ and $v_2 = 2V$, find v_o in the op amp circuit shown below.



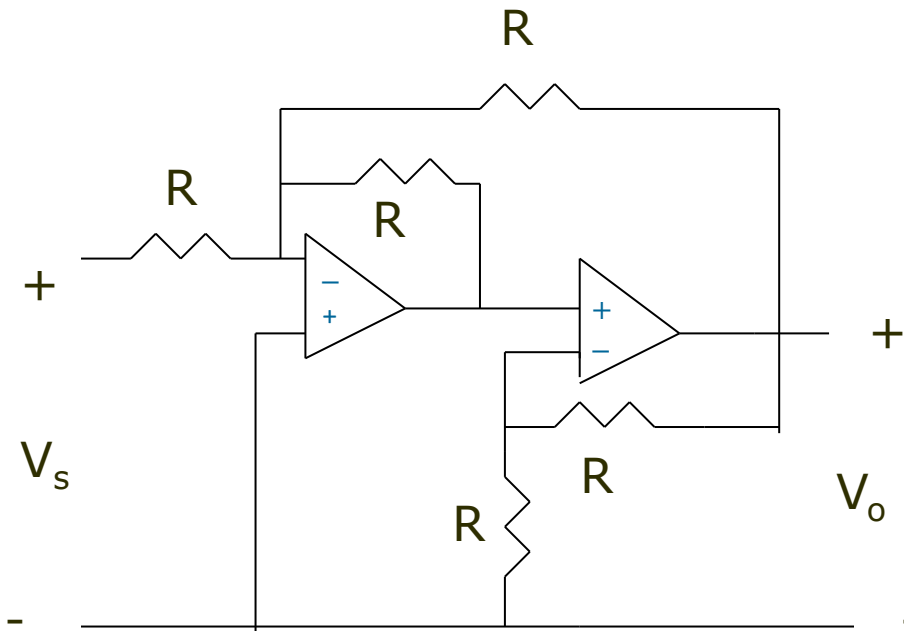
*Refer to in-class illustration, textbook

Ans: 8.667 V

5.4 Cascaded Op Amp (3)

Example

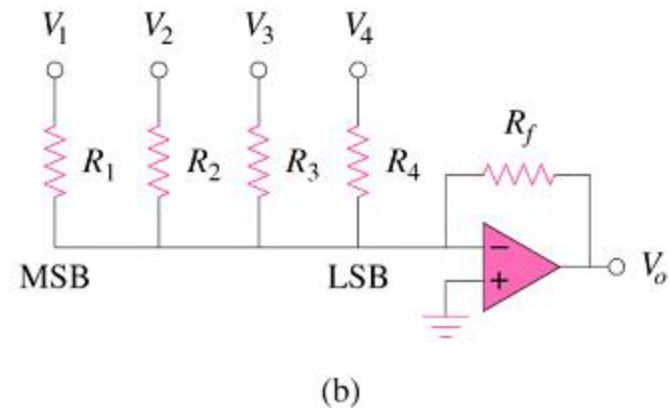
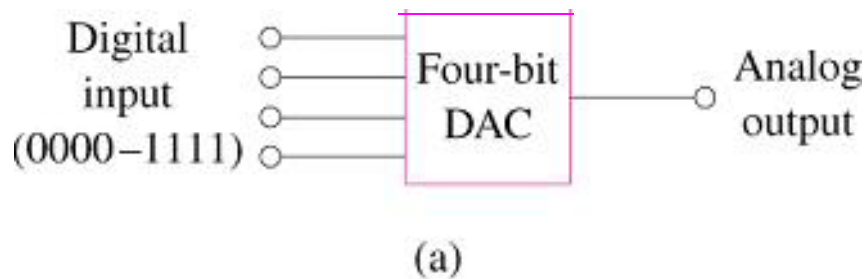
Determine the voltage transfer ratio v_o/v_s in the op amp circuit below, where $R = 10 \text{ k}\Omega$.



5.5 Application (1)

- Digital-to Analog Converter (DAC) : it is a device which transforms digital signals into analog form.

Four-bit DCA: (a) block diagram (b) binary weighted ladder type



$$-V_o = \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 + \frac{R_f}{R_4} V_4$$

where

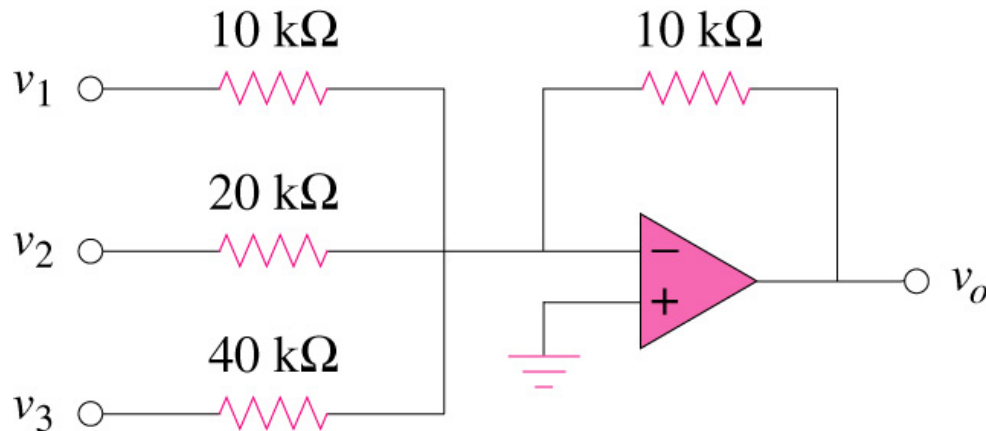
V_1 – MSB, V_4 – LSB

V_1 to V_4 are either 0 or 1 V

5.5 Application(2)

Example 8

For the circuit shown below, calculate v_o if $v_1 = 0V$, $v_2 = 1V$ and $v_3 = 1V$.



*Refer to in-class illustration, textbook

Ans: $v_o = -0.75V$